Name_____ Date_____ Period_____

LESSON 6-5 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Writing Algebraic Expressions

Maryland Standard 6.EE.A.2a · Learning target – I can write an algebraic expression that uses a variable for an unknown amount.

REMEMBER

- A **variable** is a letter that stands for an unknown amount. $5n$ means $5 \times n$.
- In $5n + 12$, 5 is the **coefficient** and 12 is the **constant**.
- Order matters: "4 less than x" is $x - 4$. An amount that repeats goes with the variable.

WORKED EXAMPLE

Bowling costs \$5 per game plus \$4 for shoes. Write an expression for the cost of g games.

Step 1 Part that repeats: \$5 for each game, $5g$

Step 2 One-time part: 4

Step 3 Join them: $5g + 4$

Answer **$5g + 4$ dollars**

WORD BANK

variable – a letter for a number you do not know

coefficient – the number in front of the letter: 5 in 5n

constant – a number alone that does not change

less than – subtract, and flip the order: 8 less than y is y – 8

PRACTICE

show your work · label units

1. Write an algebraic expression for each phrase.

Step 1 Find the operation word: increased, less, product, divided.

Step 2 Put the numbers in the right order.

a. a number n increased by 9 _____

b. 8 less than y _____

c. the product of 6 and k _____

d. m divided by 5 _____

e. 3 more than twice x _____

f. 7 times the sum of a and 2 _____

2. **Music Lessons.** Guitar lessons cost **\$15 per lesson** plus **\$20** for a book. Let L be the number of lessons. Write an expression for the **total cost**.

Step 1 The \$15 repeats, so it goes with L.

Step 2 Add the \$20 one time.

Expression_____

3. Write an expression for each situation.

a. 4 friends share p pretzels equally.
Pretzels each friend gets: _____

b. Tyrell had 50 stickers and gave away s.
Stickers left: _____

4. Pizza Party. Each pizza costs $11. The class orders p pizzas and pays a $5 delivery fee. Write an expression for the total cost. Then find the cost of 4 pizzas.

Step 1 Write the expression: the $11 goes with p, then add $5.

Step 2 Put 4 in for p and find the cost.

Show your work.

Expression and cost_____

5. Find the Error. Tickets cost $7 each, plus one $3 fee. Dmitri wrote $7 + 3n$ for the cost of n tickets. Explain his mistake. Write the correct expression.

Frame *Dmitri put the letter with ___. The $7 repeats, so the correct expression is ___.*

_____

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Tell someone at home what the variable, coefficient, and constant mean in $15L + 20$.

Name_____ Date_____ Period_____

LESSON 6-5 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Evaluating Algebraic Expressions

Maryland Standard 6.EE.A.2c · Learning target – I can evaluate an algebraic expression by substituting a value for the variable.

REMEMBER

- To **evaluate**, substitute the value, then use the order of operations.
- $4x$ means $4 \times x$. When $x = 3$, $4x = 12$, not 43.
- Powers come before multiplying: $2n^2$ with $n = 5$ is $2 \times 25 = 50$.

WORKED EXAMPLE

Evaluate $3a^2 - 4$ when $a = 4$.

Step 1 Substitute: $3 \times 4^2 - 4$

Step 2 Power, multiply, subtract: $3 \times 16 - 4 = 48 - 4 = 44$

Answer **44**

WORD BANK

substitute – put the number in place of the letter

evaluate – find the value

4x – 4 × x (multiply)

PRACTICE

show your work · label units

1. Evaluate each expression when $x = 6$.

Step 1 Put 6 in place of x.

Step 2 Do powers, then × and ÷, then + and −.

a. $5x =$ ____ **b.** $x + 17 =$ ____ **c.** $40 - 3x =$ ____

d. $x^2 + 4 =$ ____ **e.** $x \div 2 + 9 =$ ____ **f.** $2(x - 1) =$ ____

2. Evaluate each expression when $a = 4$ and $b = 5$.

a. $3a + 2b =$ ____ **b.** $a^2 - b =$ ____

3. Kayak rental costs $8h + 5$ dollars for h hours. Complete the table.

Step 1 Put each number of hours in for h.

Step 2 Multiply by 8, then add 5.

Hours, *h*	**1**	**3**	**6**
Cost, $8h + 5$ (dollars)			

4. Phone Battery. A phone has **100%** battery. The expression $100 - 6h$ gives the percent **left** after h hours. Find the percent left after **5 hours**.

Step 1 Multiply 6×5 first.

Step 2 Subtract the answer from 100.

Show your work.

Answer_____ %

5. Find the Error. Sam says $20 - 3k = 68$ when $k = 4$. What is the correct value?

Frame *The correct value is ___ because I multiply 3 × 4 ___ I subtract.*

☐ 68 ☐ 8 ☐ 32

TEST PRACTICE

choose the best answer

6. What is the value of $5n - 7$ when $n = 3$?

A 46 **B** 22 **C** 8 **D** 15

7. Tickets cost \$9 each and parking costs \$6. Which expression gives the total cost for t tickets?

A $6t + 9$ **B** $15t$ **C** $9(t + 6)$ **D** $9t + 6$

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___